SATHWIKA RAJ BANDARU

AI & ML Engineer

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Profile

Master's student in AI Systems at EPITA, focused on building and deploying end-to-end AI solutions for real-world use cases. Especially interested in leveraging LLMs and agentic AI to improve workflows and connect technical systems with business impact.

Education

EPITA	Master's - Artificial Intelligence Systems	2025 – Present
Kalasalingam University of Technology	Bachelor's of Technology - AIML	2021 – 2025
Politechnika Krakowska	Semester Exchange - AIML	Feb 2024

Technical Skills

Programming : Python, JavaScript
Artificial Intelligence : Machine Learning, Deep Learning, NLP, Computer Vision
Frameworks : PyTorch, Scikit-learn, FastAPI, Streamlit
Data & Tools : Pandas, NumPy, PostgreSQL, Apache Airflow
MLOps : Docker, Git, REST APIs
Visualization & Deployment : Vercel

Projects

LLM based RAG + Agentic AI System | *LLMs, FAISS, Multi-Agent* *2025*
- Built a RAG-based GenAI chatbot using LLMs and FAISS for document-grounded Q&A.
- Developed a multi-agent pipeline improving response relevance and reliability.

Real Time Bike Demand Forecasting | *Airflow, PostgreSQL* *2024*
- Developed a containerized machine learning pipeline using Apache Airflow, Docker, and PostgreSQL, automating data processing, model training, and deployment of a REST API for real-time predictions.

Clinical Text Intelligence System | *NLP, Rest API* *2024*
- Built an AI application for medical text analysis with entity extraction, sensitive data protection, and automated summarization using FastAPI and Streamlit, achieving an F1-score of **0.92**.

Medical Image Diagnosis System | *Deep Learning, Grad-CAM* *2024*
- Designed a CNN-based medical image diagnosis system with visual interpretability using Grad-CAM, achieving **97%** accuracy and **95.4%** recall.

Disease Risk Prediction System | *Tabular ML* *2024*
- Developed a machine learning model to predict disease risks (diabetes, heart disease, stroke) from clinical data, incorporating interpretability with SHAP and achieving an AUC-ROC of **0.89**.

Publications

Object Identification for Visually Impaired using IoT with Image Processing
- Published research on assistive technology for visually impaired individuals using IoT and computer vision.

Soft Skills & Languages

Soft Skills :Teamwork, Technical Communication, Problem Solving, Analytical Thinking
Languages : English (Fluent), French (En cours), Telugu (Mother tongue), Hindi (Fluent)